

Fig. 2: Actual-size, single-side PCB for the circuit

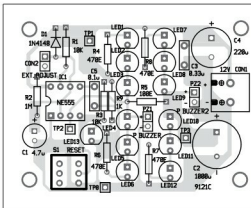


Fig. 3: Component layout for the PCB

can be in the range of 4.5V to 15V as well, depending on the number of LEDs used in each branch. Higher number of LEDs will require higher voltage. LED13 glows when power supply is connected in the circuit.

Resistors R4, R6, R7 and R8 are selected according to the number and type of the LEDs used. If the values of these resistors are too low, the output of the timer will be overloaded and the LEDs in the upper and the lower branches will

get activated simultaneously.

Overloading may also damage the NE555 timer. It is suggested to keep the total output current drawn from NE555 below 100mA.

On/off switch S1 is used to start or stop the alarm. Connector CON2 is an optional input point for connecting a variable element, such as a preset, for adjusting or varying the frequency of the square signal for more audio-visual effects.

Connect piezo buzzers PZ1 and PZ2 at their provided places in the PCB. Also connect 2-pin terminal CON1 for power supply. Connect CON2 for external input (optional). Before using the alarm circuit, check at the test points given in the table. ●

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